Schur Covariance Evaluation

A Principled Pseudo-Likelihood in High Dimensions

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Abstract

The Gaussian log-likelihood is the standard criterion for scoring a covariance estimate, but in high dimensions it is governed by the smallest eigenvalues of the estimate—precisely the part of the spectrum that is unidentifiable when the dimension is comparable to the sample size—and we show that it ranks estimators below chance. We introduce the *Schur pseudo-likelihood*, a one-parameter family ℓ_γ, $\gamma \in [0, 1]$, obtained by damping the cross-block coupling of the Gaussian likelihood through its Schur complements. At $\gamma = 1$ the family is the exact joint likelihood; at $\gamma = 0$ it is the block-diagonal (composite) likelihood; intermediate γ yields a better-conditioned interpolant. The construction is the evaluation-side counterpart of Schur complementary portfolio allocation, which interpolates hierarchical risk parity and the minimum-variance portfolio through the same Schur-complement control. In a tractable two-block model we obtain the optimal coupling strength in closed form—it is the *reliability* of the coupling, a Wiener/James–Stein shrinkage—and show that the maximizer of ℓ_γ recovers the true predictive law for every γ while implying an inflated covariance, which clarifies why ℓ_γ is a scoring and regularization device rather than a free estimation objective. We characterize when an interior γ improves discrimination, locate the high-dimensional advantage in the variance of the estimate rather than the test sample, and state precisely what the present analysis does and does not establish.

1 The high-dimensional failure of the Gaussian likelihood

The conventional criterion for the quality of a covariance estimate is the likelihood it assigns to held-out data. For a zero-mean Gaussian,

$$\ell(\widehat{\Sigma}; x) \;=\; -\tfrac{1}{2}\big(\log\det\widehat{\Sigma} + x^\top\widehat{\Sigma}^{-1}x\big),$$

or, in terms of the sample covariance S of a test set, $\ell(\widehat{\Sigma}; S) = -\frac{1}{2}\big(\log\det\widehat{\Sigma} + \operatorname{tr}(\widehat{\Sigma}^{-1}S)\big)$, which is proportional to the Wishart log-likelihood and, up to constants, to the negative Stein loss [James and Stein, 1961]. The criterion is a strictly proper scoring rule [Gneiting and Raftery, 2007] and is the objective that maximum likelihood optimizes; in low dimensions it is the appropriate choice.

Both terms are, however, controlled by the smallest eigenvalues of $\widehat{\Sigma}$: the log-determinant is $\sum_i \log\lambda_i$, which diverges to $-\infty$ as any $\lambda_i \to 0$, and $\widehat{\Sigma}^{-1}$ diverges in the same limit. When the dimension p is comparable to the sample size n, these eigenvalues concentrate near zero and become statistically unidentifiable (the Marčenko–Pastur regime), so the likelihood is determined largely by the least reliable part of the spectrum. Empirically the effect is pronounced. In a controlled experiment in which the correct ordering of two estimates is known—the true Kullback–Leibler divergence is available because the population covariance is specified—the held-out likelihood selects the better estimate in approximately 45% of trials, below the chance rate, whereas inversion-free criteria such as the Frobenius distance and the variogram score [Scheuerer and Hamill, 2015] attain approximately 85%. This is the evaluation-side analogue of the instability of the minimum-variance portfolio $w \propto \widehat{\Sigma}^{-1}\mathbf{1}$, which fails through the same $\widehat{\Sigma}^{-1}$ and the same eigenvalues. In both settings the remedy is to discount the raw inverse; the contribution of this paper is to do so continuously, through a single interpretable parameter, rather than by discarding the cross-block structure outright.

2 Exact Schur factorization

Partition the variables into K ordered blocks, $x = (x_1, \dots, x_K)$. The joint density factors exactly into a product of block conditionals, $p(x) = \prod_k p(x_k \mid x_{<k})$, and for a Gaussian each conditional is Gaussian with covariance the Schur complement

$$S_k \;=\; \Sigma_{kk} - \Sigma_{k,<k}\, \Sigma_{<k,<k}^{-1}\, \Sigma_{<k,k}$$

and mean $\mu_{k|<k} = \Sigma_{k,<k}\Sigma_{<k,<k}^{-1}x_{<k}$. Consequently $\log\det\Sigma = \sum_k \log\det S_k$, the quadratic form decomposes blockwise, and the likelihood is a sum of block-conditional terms, each of which inverts only the conditioning block $\Sigma_{<k,<k}$. This is the block Cholesky factorization, and it underlies the Vecchia approximation [Vecchia, 1988, Katzfuss and Guinness, 2021, Pan et al., 2024], Gaussian–Markov-random-field likelihoods [Rue and Held, 2005], and nested-dissection elimination [George, 1973]. No approximation has been introduced; the likelihood has merely been rewritten in a form amenable to a single modification.

3 The Schur pseudo-likelihood

We damp the Schur complement, and the associated conditional regression, by a parameter $\gamma \in [0, 1]$:

$$S_k(\gamma) \;=\; \Sigma_{kk} - \gamma\, \Sigma_{k,<k}\Sigma_{<k,<k}^{-1}\Sigma_{<k,k} \;=\; (1-\gamma)\,\Sigma_{kk} + \gamma\, S_k, \qquad \mu_{k|<k}(\gamma) = \gamma\,\Sigma_{k,<k}\Sigma_{<k,<k}^{-1}x_{<k},$$

and define the score $\ell_\gamma(\Sigma; x) = \sum_k \log \mathcal{N}\big(x_k;\, \mu_{k|<k}(\gamma),\, S_k(\gamma)\big)$. At $\gamma = 1$ this is the exact joint likelihood. At $\gamma = 0$ each block is scored under its unconditional covariance Σ_{kk} with no reference to the others, which is the block-marginal composite likelihood [Besag, 1975, Lindsay, 1988, Varin et al., 2011]. For $\gamma \in (0, 1)$ the family modifies the block covariances rather than the data, inverts no matrix larger than a conditioning block, and is better conditioned than the full likelihood because damping raises the small eigenvalues. We refer to ℓ_γ as the Schur pseudo-likelihood.

The nature of the object requires care, as it bears on the analysis below. At $\gamma = 1$, ℓ_γ is a normalized joint density; at $\gamma = 0$ it is a composite likelihood, a sum of valid component log-densities that does not integrate to one. For $\gamma \in (0, 1)$ each term remains a proper Gaussian log-density on its block, but the damped conditionals are mutually inconsistent—they are not the conditionals of any single joint Gaussian—so ℓ_γ is a pseudo-likelihood. As shown in the next section, ℓ_γ is therefore appropriate for scoring a covariance, or for damping a given one, but not for free maximization over Σ.

4 Behavior in a tractable model

The action of γ can be characterized exactly in the smallest informative case. Let $x_1 \sim \mathcal{N}(0, a)$ and $x_2 = b\,x_1 + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, s)$, so that the true conditional law of x_2 given x_1 has slope b and conditional variance s.

Proposition 1 (Two-block model)**.** *For every $\gamma \in (0, 1]$ the maximizer of the population ℓ_γ recovers the true predictive law—the effective slope equals b and the damped conditional variance equals s—while the implied covariance is inflated, with off-diagonal $\widehat{\Sigma}_{12} = \Sigma^\star_{12}/\gamma$. The implied covariance is positive definite if and only if $\gamma^2(1-\rho^2) > (1-\gamma)\rho^2$, where $\rho^2 = b^2a/(b^2a + s)$ is the conditional R^2; below this threshold the free maximizer leaves the positive-definite cone.*

The inflation is the key qualitative fact: a free fit compensates for the damping by scaling the coupling by $1/\gamma$, and for sufficiently strong coupling this scaling produces a non-positive matrix. This is the precise sense in which ℓ_γ fails as a free estimation objective for $\gamma < 1$, and it is consistent with use as a score or as a fixed transform, both of which hold $\widehat{\Sigma}$ fixed.

The quantity that γ controls—the degree of trust placed in the estimated coupling—admits a well-defined optimum. Estimating the slope from n observations by least squares gives an unbiased $\hat{b}$ with variance $s/(a(n-2))$. Minimizing the expected predictive residual over γ yields

$$\gamma^\star = \frac{b^2}{b^2 + \operatorname{Var}\hat{b}} = \frac{(n-2)\,\rho^2}{(n-2)\,\rho^2 + (1-\rho^2)}.$$

Thus $\gamma^\star$ is the reliability of the coupling, a Wiener/James–Stein shrinkage: it increases toward 1 as the sample size or coupling strength grows and decreases toward 0 as the coupling becomes uninformative, and is interior when the coupling is partially reliable. The closed form agrees with the simulated optimum. In the vector case ρ^2 is replaced by the largest squared canonical correlation between the blocks; across K blocks the inflation, and the positive-definiteness threshold, compound along the conditioning chain.

5 Discrimination efficiency and the Godambe boundary

It is tempting to credit the interior optimum to the variance of the held-out score, but that attribution is mistaken, and separating the two sources of variability is what makes the mechanism clear. Write the score as a quadratic form, $\ell_\gamma(\widehat{\Sigma}; x) = -\frac{1}{2}\big(\mathrm{Ld}_\gamma + x^\top W_\gamma(\widehat{\Sigma})\, x\big)$, where W_γ assembles the damped block conditionals and $W_1 = \widehat{\Sigma}^{-1}$. For two fixed estimates and a Gaussian test sample, the probability of ranking the better estimate above the worse is governed by a per-sample signal-to-noise ratio, which is maximized at $\gamma = 1$; under test-sample noise alone the full likelihood is optimal, and damping introduces only bias. The high-dimensional advantage of an interior γ is therefore not attributable to test-sample variance.

The advantage arises from the variability of the estimate. The score depends on $\widehat{\Sigma}$ through W_γ, and at $\gamma = 1$ the sensitivity of $W_1 = \widehat{\Sigma}^{-1}$ scales as $1/\lambda_{\min}^2$; the variance of the score difference across training draws then scales as $1/\lambda_{\min}^4$, which grows sharply as the population becomes ill-conditioned, whereas a damped γ keeps it bounded. When the discriminating signal resides in the within-block structure and the cross-block coupling is shared noise, an interior γ strictly improves on both endpoints (a ranking accuracy of approximately 0.84 against 0.45 at the extremes). The parameter is thus a bias–variance control over the estimate rather than over the test point.

Finally, the estimating equation $\nabla_\Sigma \ell_\gamma$ is unbiased at the truth only at the endpoints—$\gamma = 1$ (the Fisher score) and $\gamma = 0$ (correctly specified block marginals)—and is biased in the interior, reflecting the inflation of the previous section. Classical Godambe efficiency therefore applies only at the endpoints, where $\gamma = 0$ carries the standard composite-likelihood efficiency loss relative to the full likelihood; the interior is a tempering, justified by discrimination rather than by consistency. This formalizes the distinction between scoring with ℓ_γ and estimating with it.

6 Damping as structured shrinkage

The transform interpretation—damping the coupling of a given covariance—is a shrinkage. Each conditional covariance is shrunk from its full Schur complement toward the unconditional block, $S_k(\gamma) = (1-\gamma)\Sigma_{kk} + \gamma S_k$, with intensity $1-\gamma$. Since conditioning reduces variance, $S_k \preceq \Sigma_{kk}$, this raises the conditional covariance and lifts the small eigenvalues of every block-sized inverse the score performs; it is a James–Stein shrinkage applied blockwise to the conditioning. The operation is complementary to spectral shrinkage: Ledoit–Wolf and rotation-invariant estimators regularize the eigenvalues (basis-free), whereas Schur damping regularizes the cross-block coupling (basis-dependent). The two compose, and γ is to the block factorization what the shrinkage intensity [Ledoit and Wolf, 2004, 2012, Bun et al., 2017] is to the spectrum.

7 A unifying view

The same control organizes several constructions onto two axes of the block factorization. The first axis is the conditioning structure—which earlier blocks each block conditions on—which the Vecchia approximation, Gaussian–Markov random fields, banding, and nested dissection regularize through sparsity at full coupling strength. The second axis is the coupling strength γ, which those methods do not vary. The axes are orthogonal and compose.

	$\gamma = 1$ (full trust)	$\gamma = 0$ (no trust)
dense conditioning	exact Gaussian likelihood; MVO $w \propto \Sigma^{-1}\mathbf{1}$	block composite likelihood; HRP
sparse conditioning	block-Vecchia / GMRF likelihood	(block-marginal, same column)
interior γ: the Schur pseudo-likelihood, the interpolant absent from the above		

The portfolio entries are not analogical. Minimum-variance optimization is the dense-conditioning, $\gamma = 1$ corner; hierarchical risk parity [López de Prado, 2016] is the $\gamma = 0$ corner; Schur complementary allocation [Cotton, 2024] interpolates them by damping the same Schur complement through the same γ. The Schur pseudo-likelihood is that construction applied to evaluation, and the reason an interior γ improves on the minimum-variance endpoint is the reason hierarchical risk parity is more stable than unconstrained mean–variance optimization [Antonov et al., 2024].

8 Generality

The construction requires only a Gaussian (or Gaussian-pseudo) block likelihood and a partition, so the coupling-strength control applies to Gaussian-block models generally. In spatial statistics and Gaussian-process regression, where Vecchia and GMRF likelihoods condition each location on a neighborhood, γ supplies a robustness control on the strength of that conditioning, orthogonal to the choice of neighborhood. In Gaussian graphical models it tempers conditional-dependence strength between the full penalized likelihood and node-wise pseudo-likelihood. In longitudinal and state-space models, with time slices as blocks, it interpolates between independent slices and the full temporal joint. Wherever composite likelihood is used for tractability, γ provides a continuous path back toward the full likelihood. These are statements about the shared mechanism rather than empirical results; the covariance-evaluation case is the one studied directly here.

9 Interpolation geometry and empirical results

Damping interpolates each conditional covariance between Σ_{kk} and S_k along a straight line. An alternative is the affine-invariant SPD geodesic, $\Sigma_{\mathrm{BD}} \#_\gamma \Sigma = \Sigma_{\mathrm{BD}}^{1/2}(\Sigma_{\mathrm{BD}}^{-1/2}\Sigma\,\Sigma_{\mathrm{BD}}^{-1/2})^\gamma \Sigma_{\mathrm{BD}}^{1/2}$, which lifts small eigenvalues multiplicatively. The two interpolations agree at the endpoints and to first order and differ near a singular Schur complement; in that strong-coupling regime the linear interpolation exhibits a loss of discriminating power that the geodesic interpolation does not.

The empirical findings are threefold. The optimal γ is regime-dependent: it approaches 1 when the coupling is reliable, approaches 0 when it is noise, and is strictly interior—improving on both endpoints—when the coupling is partially reliable, consistent with the bias–variance account above. The geodesic variant is the more stable default where the two interpolations diverge. Finally, on real equity returns (Ken French portfolios, rolling minimum-variance out-of-sample variance), estimators that invert an unregularized covariance degrade by orders of magnitude while well-conditioned shrinkage does not; this is not a test of ℓ_γ as an equity estimator, but it corroborates the central mechanism, namely that high-dimensional performance is governed by the conditioning of the inverse.

10 Limitations

The closed-form $\gamma^\star$ is established for the predictive (estimator) use in the two-block model. The high-dimensional discrimination optimum is only partially resolved: the variance term is first-order and is given above ($1/\lambda_{\min}^4$ at $\gamma = 1$, bounded below it), but the signal term is second-order in the sampling noise, so a closed form for the discrimination $\gamma^\star$ requires a higher-order analysis and is at present available only numerically. The compounded K-block positive-definiteness threshold is established as a two-block formula together with numerical evidence of compounding, not yet as a closed-form chain. The dependence on block ordering, as in Vecchia approximations, is not addressed here. The central claim is falsifiable: the interior optimum should disappear when the coupling is either fully reliable or pure noise, and an interior optimum in a regime known to be fully reliable would contradict it.

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